Math 591: Differentiable Manifold

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Abstract

Math 591 taught be Professor Alejandro Uribe.

Contents

Chapter 1

Point Set Topology and differentiable manifold

Lecture 1: Point set topology

1.1 Review of point set topology

31 Aug. 9:00

Definition 1.1.1. Let X be a topological space. A basis for X is a collection $(U_\alpha)_{\alpha\in A}$ of open sets in X such that, for any $p \in X$, and open set U, $p \in U$, there is $\alpha \in A$ such that $p \in U_\alpha \subseteq U$.

Claim. This is equivalent to: for any open set $U \subset X$, there is a subset $A_U \subseteq A$ such that $U = \bigcup_{\alpha\in A_U} U_\alpha$.

Definition 1.1.2. A neighborhood of a point $p \in X$ is any open set U such that $p \in U$.

Definition 1.1.3. A basis of neighborhoods of a point $p \in X$ is a collection $\{V_\alpha\}_\alpha \in A_p$ such that for any neighborhood U of p, there is $\alpha \in A_p$ such that $V_\alpha \subseteq U$.

Definition 1.1.4. Let X be a topological space. X is called *second countable* if there is a countable basis $\{U_\alpha\}_{\alpha\in A}$ for X.

Example. Let $X = \mathbb{R}^n$ with the standard topology. A countable basis is $\{B_{\alpha,r}\}_{\alpha\in\mathbb{Q}^n, r\in\mathbb{Q}^+}$, where $B_{\alpha,r} = \{x \in \mathbb{R}^n : |x-\alpha| < r\}$.

Example (Non-example)**.** $X = \mathbb{R}$ with the discrete topology is not second countable, since for any $x \in \mathbb{R}$ we need a basis.

Definition 1.1.5. X is *Hausdorff (or T_2)* if for any $p, q \in X$, there are neighborhoods U of p and V of q such that $U \cap V = \emptyset$.

Example. Metric spaces are Hausdorff.

Example (Non-example)**.** The real line with 2 origins is not Hausdorff. As a set, $X = (-\infty, 0) \cup \{0_1, 0_2\} \cup (0, \infty)$. The topology is generated by open sets in $(-\infty, 0)$ and $(0, \infty)$, plus basis of neighborhoods of 0_1 and 0_2: $\{(-\epsilon, 0) \cup \{0_i\} \cup (0, \epsilon) : \epsilon > 0\}$, $i = 1, 2$. Then it is easy to see that 0_1 and 0_2 cannot be separated by neighborhoods.

Definition 1.1.6. A *topological manifold* is a topological space that is

1) Second countable.
2) Hausdorff.
3) Locally Euclidean.

Definition 1.1.7. Let X be a topological space. We say X is *locally Euclidean* if for any $p \in X$, there is a neighborhood U of p and a homeomorphism $\phi : U \to \mathbb{R}^n$.

Remark.
- Some books allow n to depend on p, but then n will be constant on a connected component.
- In the definition, we can replace $\mathbb{R}^n$ by any open subset $A \subset \mathbb{R}^n$. One side is clear. For the other side, let B be an open ball of p contained inside A, then we can compose $\phi|_{\phi^{-1}(B)}$ with a homeomorphism from B to $\mathbb{R}^n$.

Remark. The 3 conditions in the definition of a topological manifold are independent.

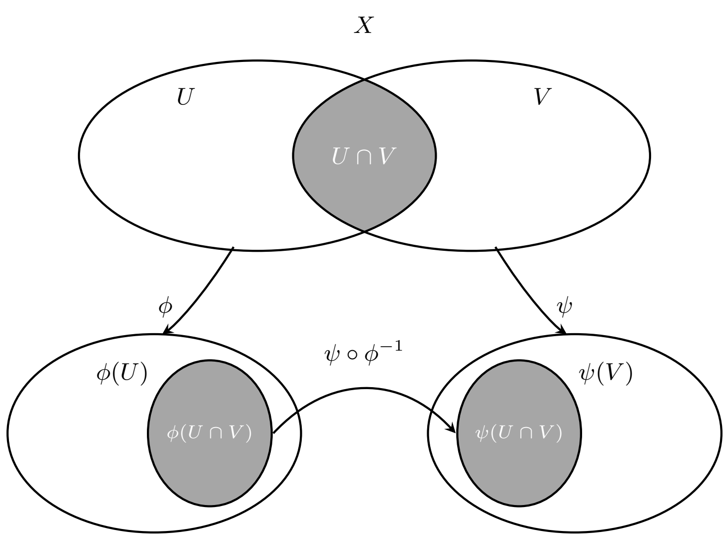
- Line with 2 origins is locally Euclidean and second countable, but not Hausdorff. To see locally Euclidean, note we have a homeomorphism from a neighborhood of 0_1: $\phi : (-\epsilon, 0) \cup \{0_1\} \cup (0, \epsilon) \to (-\epsilon, \epsilon)$.
- $X = \mathbb{Q}$ is Hausdorff and second countable, but not locally Euclidean.
- $\mathbb{R} \times \mathbb{R}$, where the first $\mathbb{R}$ has the discrete topology and the second $\mathbb{R}$ has the standard topology, is Hausdorff and locally Euclidean (take $\{x\} \times \mathbb{R}$), but not second countable.

Example. $\mathbb{S}^2$ with subspace topology is a topological manifold. Let $p \in \mathbb{S}^2$. WLOG, assume the z-coordinate of p is positive. Then $U = \{(x, y, z) \in \mathbb{S}^2 : z > 0\}$ is a neighborhood of p. Let $\phi : U \to \mathbb{R}^2$ be $\phi(x, y, z) = (x, y)$. Then ϕ is a homeomorphism.

Some preview:

Suppose X is a topological manifold. Take $p \in X$, and two U, V neighborhoods of p. Let

$$\begin{cases} \phi : U \to A^{\text{open}} \subseteq \mathbb{R}^n, \\ \psi : V \to B^{\text{open}} \subseteq \mathbb{R}^n \end{cases}$$



$\psi \circ \phi^{-1}$ is called a *transition map*. It is a homeomorphism from $\phi(U \cap V)$ to $\psi(U \cap V)$.

Lecture 2: Point set topology

1.2 Product Topology, Subspace Topology, and Quotient Topology

31 Aug. 9:00

Definition 1.2.1. Let X and Y be topological spaces. A subset $A \subseteq X \times Y$ is open if and only if for any $(x, y) \in A$, there are open sets $U \subseteq X$ and $V \subseteq Y$ such that $(x, y) \in U \times V \subseteq A$.

Remark. In categorical language, $X_1 \times X_2$ is the object in the category of topological spaces that satisfies the following universal property: for any topological space Y and continuous maps $f_1 : Y \to X_1$, $f_2 : Y \to X_2$, there is a unique continuous map $f : Y \to X_1 \times X_2$ such that $\pi_1 \circ f = f_1$ and $\pi_2 \circ f = f_2$, where $\pi_i : X_1 \times X_2 \to X_i$ is the projection map.

$$\begin{tikzcd} & Y \arrow[dl, "f_1"'] \arrow[d, dashed, "\exists!\, f"] \arrow[dr, "f_2"] & \\ X_1 & X_1 \times X_2 \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & X_2. \end{tikzcd}$$

Definition 1.2.2. Let X be a topological space, and let $\sim$ be an equivalence relation on X. Let $\pi : X \to X/\sim$ be the quotient map. The quotient topology on $X/\sim$ is defined as follows: a subset $U \subseteq X/\sim$ is open if and only if $\pi^{-1}(U)$ is open in X.

Remark. If X is Hausdorff or second countable, then $X/\sim$ may not be Hausdorff or second countable.

Proposition 1.2.1. Let Y be a topological space, and let $f : X/\sim \to Y$ be a map. Then f is continuous if and only if $f \circ \pi : X \to Y$ is continuous.

Proof. "$\Rightarrow$" is trivial, since π is continuous.
"$\Leftarrow$" Let $U \subseteq Y$ be open. Then $(f \circ \pi)^{-1}(U)$ is open in X. But $(f \circ \pi)^{-1}(U) = \pi^{-1}(f^{-1}(U))$. So $f^{-1}(U)$ is open in $X/\sim$. ■

Definition 1.2.3. An equivalence relation $\sim$ on a set X is open if and only if the quotient map $q : X \to X/\sim$ is an open map.

Example (Non-example). Take $X = \mathbb{R}$, and define $\sim$ by setting $0 \sim 1$

Note. $\sim$ is open if and only if for any open set $U \subseteq X$, $\pi^{-1}(\pi(U))$ is open in X.

Definition 1.2.4. $\pi^{-1}(\pi(U)) = \{x \in X : x \sim y \text{ for some } y \in U\}$ is called the *saturation* of U.

Theorem 1.2.1. Assume $\sim$ is an open relation in X. Define $\Gamma = \{(x, y) \in X \times X \mid x \sim y\}$. Then $X/\sim$ is Hausdorff if and only if Γ is closed in $X \times X$.

Remark. Suppose $\sim$ is trivial, i.e. $x \sim y$ if and only if $x = y$. Then $\Gamma = \{(x, x) \mid x \in X\}$ is the diagonal of $X \times X$, and $X/\sim = X$. So the theorem says that X is Hausdorff if and only if the diagonal is closed in $X \times X$.

Proof. "$\Leftarrow$" Assume that Γ is closed. Let $x, y \in X$ such that $x \not\sim y$. Then $(x, y) \notin \Gamma$. Since Γ is closed, there are open sets $U, V \subseteq X$ such that $(x, y) \in U \times V$ and $U \times V \cap \Gamma = \emptyset$. Then $\pi(U)$ is open, because $\sim$ is an open relation. Similarly, $\pi(V)$ is open.

Claim. $\pi(U) \cap \pi(V) = \emptyset$.

Proof. If not, then there are $a \in U$, $b \in V$ such that $\pi(a) = \pi(b)$, i.e. $a \sim b$. Then $(a, b) \in U \times V \cap \Gamma$, which is a contradiction. ∎

Conversely, assume that $X/\sim$ is Hausdorff. We want to show that Γ is closed. Let $(x, y) \in (X \times X) \setminus \Gamma$, i.e. $x \not\sim y$. So $\pi(x) \neq \pi(y)$. By assumption, there are open neighborhoods A of $\pi(x)$ and B of $\pi(y)$ such that $A \cap B = \emptyset$.

Claim. $(\pi^{-1}(A) \times \pi^{-1}(B)) \cap \Gamma = \varnothing$.

Proof. Suppose there is $(a, b) \in \Gamma$ and $a \in \pi^{-1}(A)$, $b \in \pi^{-1}(B)$. Then $\pi(a) \in A$ and $\pi(b) \in B$. But $\pi(a) = \pi(b)$, which is a contradiction. ∎

∎

Theorem 1.2.2. If $\sim$ is an open relation in X, and X is second countable, then $X/\sim$ is second countable.

Proof. Project a countable basis. ∎

Note (Application). Take $\mathbb{S}^{2n+1} \subseteq \mathbb{R}^{2n+2} \cong \mathbb{C}^{n+1}$, take $\mathbb{S}^1 = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$. Define $\sim$ on S^{2n+1} by $z \sim w$ if and only if $z = \zeta w$ for some $\zeta \in \mathbb{S}^1$.

Definition 1.2.5. $\mathbb{S}^{2n+1}/\sim = \mathbb{CP}^n$ is called the *complex projective space.*

Claim. The quotient topology on $\mathbb{CP}^n$ is Hausdorff.

Proof. We first prove the following claim:

Claim. $\sim$ is an open relation.

Proof. Let $U \subseteq X$ be open, then $\pi^{-1}(\pi(U)) = \bigcup_{\zeta \in \mathbb{S}^1} \zeta U$. Since $x \to \zeta x$ is a homeomorphism, ζU is open. So $\pi^{-1}(\pi(U))$ is open. ∎

Consider $F : \mathbb{S}^1 \times \mathbb{S}^{2n+1} \to \mathbb{S}^{2n+1} \times \mathbb{S}^{2n+1}$ by $(\zeta, z) \mapsto (z, z\zeta)$. Since $\mathbb{S}^1 \times \mathbb{S}^{2n+1}$ is compact, and F is continuous, its image $F(S^1 \times X)$ is compact, hence closed, since $\mathbb{S}^{2n+1} \times \mathbb{S}^{2n+1}$ is Hausdorff. Take $[x], [y] \in \mathbb{CP}^2$ such that $[x] \neq [y]$. Then $(x, y) \notin F(\mathbb{S}^1 \times \mathbb{S}^{2n+1})$, so there are open neighborhoods U of x and V of y such that $(U \times V) \cap F(\mathbb{S}^1 \times \mathbb{S}^{2n+1}) = \emptyset$. Since π is open, $\pi(U)$ and $\pi(V)$ are open neighborhoods of $[x]$ and $[y]$, respectively. We claim that $\pi(U) \cap \pi(V) = \emptyset$. If not, then there are $a \in U$ and $b \in V$ such that $[a] = [b]$, i.e. $a = \zeta b$ for some $\zeta \in \mathbb{S}^1$. Then $(a, b) \in (U \times V) \cap F(\mathbb{S}^1 \times \mathbb{S}^{2n+1})$, which is a contradiction. ∎

Lecture 3: Point set topology

1.3 Topological groups and matrix groups

31 Aug. 9:00

Definition 1.3.1. A *topological group* is a group with a topology such that

1) $G \times G \to G$, $(g, h) \to gh$ and
2) $G \to G$, $g \to g^{-1}$

are continuous

Example. Any group G with the discrete topology is a topological group. Both 1) and 2) are obviously continuous.

Classical matrix groups

Matrix groups over $\mathbb{R}$

Let $\mathrm{Mat}(n,\mathbb{R}) = \{\text{all } n \times n \text{ real matrices}\}$. Then it is easy to see that $\mathrm{Mat}(n,\mathbb{R})$ is a vector space of dimension n^2. We have a continuous determinant map $\det : \mathrm{Mat}(n,\mathbb{R}) \to \mathbb{R}$. Hence, $\det^{-1}(\mathbb{R} \setminus \{0\}) = \{\text{all invertible matrices } g \in \mathrm{Mat}(n,\mathbb{R})\}$ is open in $\mathrm{Mat} \cong \mathbb{R}^{n^2}$. We can thus always consider it as a subspace of a Euclidean space, equipped with the subspace topology.

Definition 1.3.2.

- $\mathrm{GL}(n,\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$.
- $\mathrm{SL}(n,\mathbb{R}) = \det^{-1}(1) \subseteq \mathrm{GL}(n,\mathbb{R}^n)$.
- $O(n,\mathbb{R}) = \{g \in \mathrm{GL}(n,\mathbb{R}), \text{for any } x, y \in \mathbb{R}^n, gx \cdot gy = x \cdot y\}$.
- $SO(n,\mathbb{R}) = O(n,\mathbb{R}) \cap \mathrm{SL}(n,\mathbb{R})$.

Remark. Notice that $\mathrm{GL}(n,\mathbb{R})$ is a topological group since both 1) and 2) in Definition 1.3.1 are satisfied. They are algebraic maps and are hence continuous.

Remark. $g \in O(n,\mathbb{R}) \Leftrightarrow g^{-1} = g^T$.

Matrix groups over $\mathbb{C}$

Once and for all, fix $\mathbb{R}$-linear identification $\mathbb{C}^n \cong \mathbb{R}^{2n}$. Explicitly, if $(z_1, \ldots z_n) \in \mathbb{C}^n$, and write $z_j = x_j + iy_j$, then we send $(z_1, \ldots z_n) \mapsto (x_1, y_1, \ldots x_n, y_n)$.

Definition 1.3.3.

- $\mathrm{GL}(n,\mathbb{C}) = \{\text{all invertible } n \times n \text{ complex matrices}\}$.
- $U(n) = \{g \in \mathrm{GL}(n,\mathbb{C}) \mid g^{-1} = \overline{g}^T\}$.

Note. $U(1) = \{\zeta \in \mathbb{C} \setminus \{0\} \mid \zeta^{-1} = \overline{\zeta}\} = \{\text{unit circle in } \mathbb{C}\}$.

The fact is, those groups are topological manifolds. Hausdorff and second countable are inherited. It remains to prove locally Euclidean. We focus on the case of $G = \mathrm{SL}(n,\mathbb{R}) \subset \mathbb{R}^{n^2} \xrightarrow{\det} \mathbb{R}$, where $\mathrm{SL}(n,\mathbb{R}) = \det^{-1}(1)$.

Proposition 1.3.1. $\mathrm{SL}(n,\mathbb{R})$ is locally Euclidean.

Proof. Our goal will be to compute $(\nabla \det)_g$ using chain rule.

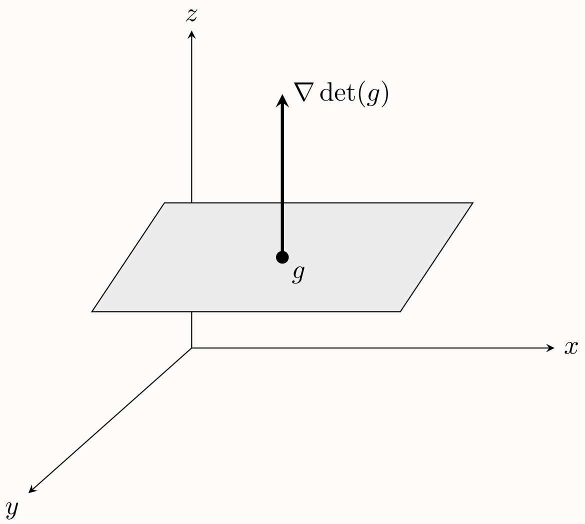
Take $g \in \mathrm{SL}(n, \mathbb{R})$, and take any $A \in \mathrm{Mat}(n, \mathbb{R})$ at g and consider $\det(g + tA)$. Now we compute

$$
\begin{aligned}
\frac{d}{dt}\big(\det(g+tA)\big)\big|_{t=0} &= \frac{d}{dt}\big(\det g \, \det\big(\mathrm{Id} + tg^{-1}A\big)\big)\big|_{t=0} \\
&\overset{V := g^{-1}A}{=} \frac{d}{dt}\big(\det(\mathrm{Id} + tV)\big)\big|_{t=0} \\
&= \frac{d}{dt}\left(\sum_{\sigma \in P_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} (\mathrm{Id} + tV)_{i,\sigma(i)}\right)\Bigg|_{t=0} \\
&= \sum_{\sigma \in P_n} \mathrm{sgn}(\sigma) \frac{d}{dt}\left(\prod_{i=1}^{n} (\mathrm{Id} + tV)_{i,\sigma(i)}\right)\Bigg|_{t=0} \\
&= \sum_{\sigma \in P_n} \mathrm{sgn}(\sigma) \frac{d}{dt}\left(\prod_{i=1}^{n} \left(\delta_{i,\sigma(i)} + tV_{i,\sigma(i)}\right)\right)\Bigg|_{t=0} \\
&= \sum_{\sigma \in P_n} \mathrm{sgn}(\sigma) \sum_{j=1}^{n} V_{j,\sigma(j)} \prod_{\substack{i=1 \\ i \neq j}}^{n} \delta_{i,\sigma(i)} \\
&= \sum_{j=1}^{n} V_{jj} \\
&= \mathrm{tr}(V) \\
&= \mathrm{tr}\big(g^{-1}A\big).
\end{aligned}
$$

We conclude that $\frac{d}{dt}(\det(g + tA))|_{t=0} = \mathrm{tr}\big(g^{-1}A\big) = (\nabla \det)_g \cdot A$. Hence for any $g \in \mathrm{SL}(n, \mathbb{R})$, we have $(\nabla \det)_g \neq \vec{0}$.

Claim. $\mathrm{SL}(n, \mathbb{R})$ is locally Euclidean.

Proof. We have the following picture:



WOLG, say the last component of $\nabla \det_g$ is nonzero. Project it to $\mathbb{R}^{n^2-1}$ by taking the first $n^2 - 1$ components. Then by implicit function theorem we see that it has a local inverse. Moreover it is locally of dimension $\dim(\mathrm{SL}(n, \mathbb{R})) = n^2 - 1$. ∎

∎

Back to general theory

Definition 1.3.4. Let G be a group and X a set. The action of G on X is a map

$$\begin{aligned} G \times X &\to X \\ (g, x) &\to g \cdot x \end{aligned}$$

such that

a) If $e \in G$ is the identity, then for any $x \in X$, $e \cdot x = x$.

b) For any $g, h \in G$, and any $x \in X$, $g \cdot (h \cdot x) = (g \cdot h) \cdot x$.

Example. All previous groups over $\mathbb{R}$ act on $X = \mathbb{R}^n$ by matrix multiplication is an example of action of G on X.

Example. IF $H \subset G$ is a subgroup, then H acts on G by group multiplication $H \times G \to G$, $(h, g) \mapsto hg$.

Definition 1.3.5. If G is a topological group, and X is a topological space, a *continuous action* is one for which $G \times X \to X$ is continuous.

Definition 1.3.6. Assume G acts on X. The *orbit* of $x \in X$ is

$$O_X := \{g \cdot x \mid g \in G\}$$

Claim. The orbits partition X.

Definition 1.3.7. An orbit relation $\sim$ is defined to be $x \sim y \Leftrightarrow$ there is $g \in G$ such that $y = g \cdot x$. The *orbit space* is $X/\sim$.

Appendix